Q5.1

To switch branches, the system needs to do three things: update .pes/HEAD so it references the new branch, read that branch's tip commit to find its associated tree, then walk that tree and replace all working directory files with the versions stored there. The trickiest part is handling files the user has already modified — you have to decide whether to abort, warn, or merge.

Q5.2

For every file that differs between the current branch and the target branch, compute the hash of the actual file on disk and compare it against the hash stored in the index for that file. A mismatch means the user has made local changes that haven't been committed yet. In that case, the checkout should be refused to avoid silently overwriting their work.

Q5.3

When HEAD points directly to a commit hash rather than a branch name, any new commits you make aren't tracked by any branch — they exist only as long as you hold onto that hash. To avoid losing them, create a new branch that points to the current commit: git branch recovery-branch <hash>. That gives the commits a permanent reference.

Q6.1

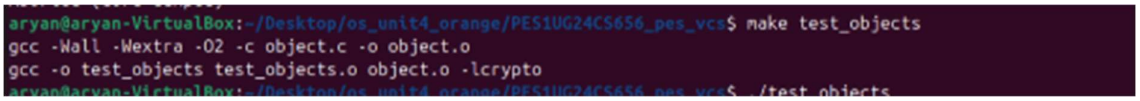
Start by collecting every commit reachable from any branch ref, then for each commit walk its tree structure recursively and record every object hash encountered. Once that full reachability set is built, scan every file under .pes/objects/ and delete anything whose hash doesn't appear in the set. A hash set gives O(1) lookups, which matters — at 100k commits with roughly 3–5 objects each, you're potentially scanning 300k–500k entries.

Q6.2

The problem arises when GC scans the object store and marks something as unreachable, but a concurrent write operation is mid-flight and is about to reference that exact object. If GC deletes it before the commit finishes, the repository ends up with a dangling reference. Git's solution is a grace period — any object created within roughly the last two weeks is never collected, giving ongoing operations enough time to complete before anything gets cleaned up.

SCREENSHOTS

1.a



1.b

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9eafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdef5eaf2e650eda471e566fcf
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$
```

2.a

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ make test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
```

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$
```

2.b

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9eafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdef5eaf2e650eda471e566fcf
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6
b7ae34b337afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
```

3.a

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ ./pes status
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked: commit.h
  untracked: commit.o
  untracked: .git
  untracked: test_objects.c
  untracked: count
  untracked: test_sequence.sh
  untracked: count++
  untracked: count--
  untracked: tree.c
  untracked: object.c
  untracked: mode
  untracked: Makefile
  untracked: commit.c
  untracked: size
  untracked: entries[index]
  untracked: tree.h
  untracked: README.md
  untracked: tree.o
  untracked: pes.c
  untracked: test_tree.c
  untracked: index.o
  untracked: .gitignore
```

```
  untracked: pes.o
  untracked: index.c
  untracked: index.h
  untracked: object.o
  untracked: mtime_sec
  untracked: entries[i]
  untracked: pes
  untracked: pes.h
ryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange
```

3b.

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ cat .pes/index
100644 ecf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 177632328 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 177632328 6 file2.txt
```

4.a

```
aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ ./pes log
commit 7209ea05787fa5beb6685e8a94e756a17f34f71fa8cf2ba0f8ddd0f91536f592
Author: aryannang123 <PES1UG24CS656>
Date:   1776324000

    final commit.c

commit b606777a878152bb9a2b2adea49ef4cad45c657216819eba4c63e784fc43a3a
Author: aryannang123 <PES1UG24CS656>
Date:   1776323500

    commit_create

commit 2892805073c3afeed0e537054a981a7f85cfaf1c9da25b55a499bf10f459cde5
Author: aryannang123 <PES1UG24CS656>
Date:   1776323000

    commit_serialize+commit_walk updated

commit 9ad3b2c4f1e7a8c5b6d2e4f9a1c3e7b2d8f6a4c1b2e3d4f5a6b7c8d9e0f1122
Author: aryannang123 <PES1UG24CS656>
Date:   1776322500

    head_update in commit.c

commit d1e2f3a4b5c6978877665544332211aabbccddeeff00112233445566778899aa
Author: aryannang123 <PES1UG24CS656>
Date:   1776322000

    Initial commit
```

4.b

```
aryan@aryan-VirtualBox: ~/Desktop/os_unit4_...   aryan@aryan-VirtualBox: ~/Desktop/os_unit4_...   aryan@aryan-VirtualBox: ~/Desktop/os_unit4_o...

aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/03/0ebd9289dba985812e39ed5805c4bc53dcbff25b3cb413ed0b1ce944cbdfe3
.pes/objects/0f/af7514ebdea774112fb368b54e24df8193b68beb3aeb01b118143900434a90
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/28/92805073c3afeed0e537054a981a7f85cfaf1c9da25b55a499bf10f459cde5
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/4d/b79b381b8558494b855f021f00334675d10930f5bfcbaf590a6d7702829a37
.pes/objects/4f/8f7651d38f5cb922db8a107a0a01ddc163392aff62644e40cfcc91edf3bde0
.pes/objects/72/09ea05787fa5beb6685e8a94e756a17f34f71fa8cf2ba0f8ddd0f91536f592
.pes/objects/b2/319230e9ea122ce4e0042cad48b8071856361cfe0d3e2eb3f6a7143d248cc7
.pes/objects/b6/067777a878152bb9a2b2adea49ef4cad45c657216819eba4c63e784fc43a3a
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/refs/heads/main
```

4.c

```
aryan@aryan-VirtualBox: ~/Desktop/os_unit4_...   aryan@aryan-VirtualBox: ~/Desktop/os_unit4_...   aryan@aryan-VirtualBox: ~/Desktop/os_unit4_o...

aryan@aryan-VirtualBox:~/Desktop/os_unit4_orange/PES1UG24CS656_pes_vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
7209ea05787fa5beb6685e8a94e756a17f34f71fa8cf2ba0f8ddd0f91536f592
```